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Canonical Divisor

Let $C = V(x^2 - yz)$ over $\mathbb{C}$.

We have the affine patches:

$$r = \frac{x}{z}, s = \frac{y}{z}$$
$$t = \frac{x}{y}, u = \frac{z}{y}$$
$$v = \frac{y}{x}, w = \frac{z}{x}$$

Compute the divisor of the differential form dr.

First from sage we have:

```
sage: K.<r> = FunctionField(QQ)
sage: _.<S> = K[]
sage: L.<s> = K.extension(r^2 - S)
sage: dr = r.differential()
sage: dr.divisor()
-2*Place (1/r)
```

which shows us $\mathrm{div}(\mathrm{d}r) = -2(0:1:0)$.

The book is wrong. We must find a local parameter/uniformizer, and write the differential as $\alpha(t)\,\mathrm{d}t$, then evaluate $\mathrm{ord}_P(\alpha(t))$.

- Canonical divisor of the projective line
- Compute principal divisor for a rational function on a curve
- Calculating the divisors of the coordinate functions on an elliptic curve
- An exposition of Riemann Roch theorem for curves

Set $z = 1$

The curve is given by $r^2 - s = 0$ but there are no poles or zeros in this patch.

Set $y = 1$

$$t^2 - u = 0 \implies \mathrm{d}u = 2t\,\mathrm{d}t$$

We note that $\frac{\partial f}{\partial t}(P) = 2t$ so u is not a local parameter for $P = (0,0)$. However t is a local parameter for all points.

$P = (0,0)$. However r is a local parameter for all points. Observe that $r = \frac{t}{u}$ and $s = \frac{1}{u}$. Using $\mathrm{d}r$ we get

$$\begin{aligned}
\mathrm{d}(\frac{t}{u}) &= \frac{1}{u}\,\mathrm{d}t + t\,\mathrm{d}(\frac{1}{u}) \\
&= \frac{1}{u}\,\mathrm{d}t - \frac{t}{u^2}\,\mathrm{d}u \\
&= \frac{1}{u}\,\mathrm{d}t - \frac{t}{u^2}(2t\,\mathrm{d}t) \\
&= \frac{u - 2t^2}{u^2}\,\mathrm{d}t \\
&= \frac{-1}{t^2}\,\mathrm{d}t
\end{aligned}$$

by substituting $\mathrm{d}t$ for $\mathrm{d}u$. Doing the opposite, we get

$$\begin{aligned}
\mathrm{d}(\frac{t}{u}) &= \frac{1}{u}\,\mathrm{d}t - \frac{t}{u^2}\,\mathrm{d}u \\
&= \frac{1}{u}\left(\frac{1}{2t}\,\mathrm{d}u\right) - \frac{t}{u^2}\,\mathrm{d}u \\
&= \left(\frac{1}{2tu} - \frac{t}{u^2}\right)\mathrm{d}u \\
&= \frac{u-2t^2}{2tu^2}\,\mathrm{d}u \\
&= \frac{-1}{2t^3}\,\mathrm{d}u \\
&= \frac{-1}{2t^3}\,\mathrm{d}(t^2) \\
&= \frac{-1}{t^2}\,\mathrm{d}t
\end{aligned}$$

While if we use $\frac{1}{2r}\,\mathrm{d}s$, we get

$$\begin{aligned}
\mathrm{d}r &= \frac{1}{2r}\,\mathrm{d}s \\
&= \frac{u}{2t}\,\mathrm{d}(\frac{1}{u}) \\
&= -\frac{u}{2tu^2}\,\mathrm{d}u \\
&= -\frac{1}{2t^3}\,\mathrm{d}u \\
&= -\frac{1}{t^2}\,\mathrm{d}(t^2)
\end{aligned}$$

so all calculations agree.

$$\operatorname{div}\left(\frac{-1}{t^2}\right) = -2\operatorname{div}(-t) = -2\operatorname{div}(t)$$

and so $\operatorname{div}(\frac{-1}{t^2}) = -2(0:1:0)$.

Set $z = 1$

$$1 - vw = 0 \Longrightarrow w\,\mathrm{d}v = -v\,\mathrm{d}w$$

We note that $\frac{\partial f}{\partial v}(P) = -w$ so w is not a local parameter for $w = 0$, however no such point exists. Likewise for v, so we can consider both of these coordinates as local.

Observe that $r = \frac{1}{w}$ and $s = \frac{v}{w}$. Using $\mathrm{d}r$ we get $\mathrm{d}r = -\frac{1}{w^2}dw$.

While if we use $\frac{1}{2r}\,\mathrm{d}s$, we get

$$\begin{aligned}
\mathrm{d}r &= \frac{1}{2r}\,\mathrm{d}s \\
&= \frac{w}{2}\,\mathrm{d}(\frac{v}{w}) \\
&= \frac{w}{2}\left(\frac{1}{w}\,\mathrm{d}v - \frac{v}{w^2}\,\mathrm{d}w\right) \\
&= \frac{w}{2}\left(\frac{1}{w}\,\mathrm{d}v - \frac{v}{w^2}\left(\frac{-w}{v}\,\mathrm{d}v\right)\right) \\
&= \mathrm{d}v \\
&= \frac{w}{2}\left(\frac{1}{w}\left(\frac{-v}{w}\,\mathrm{d}w\right) - \frac{v}{w^2}\,\mathrm{d}w\right) \\
&= \frac{w}{2}\left(\frac{1}{w}\frac{-v}{w} - \frac{v}{w^2}\right)\mathrm{d}w \\
&= \frac{w}{2}\frac{-2v}{w^2}\,\mathrm{d}w \\
&= \frac{-1}{w^2}\,\mathrm{d}w
\end{aligned}$$

There is no $w = 0$ on this patch however.